PLaID++: Symmetry-aware crystal generation with diversity-aware preference optimization

Method overview reconstructed from the ICML 2026 paper and official implementation

A Wyckoff serialization and supervised fine-tuning

MP-20 structures
CIF inputs

Symmetry analysis
Refine structure
PyXtal: SG + inequivalent sites

Compact Wyckoff text
Tb8Mg4 | Spacegroup: Pnma
abc: 7.11 5.33 9.75 | Sites (12)
Tb 0.818 0.250 0.084 4c
Tb 0.467 0.250 0.728 4c
Mg 0.247 0.250 0.124 4c

Qwen2.5-7B + LoRA
Supervised fine-tuning

Generation ~2/3
Element infill ~1/3

Initial policy π_0
Autoregressive model

1 Symmetry-informed text encoding

Representative coordinate + Wyckoff label per orbit; symmetry expands equivalent sites.

B Iterative RLIP / DPO preference optimization

2 Diversity-aware RLIP

Stability tiers + novelty preferences; rising sampling temperature helps resist mode collapse.

3 Joint conditional + unconditional training

Policy π_{i-1}
Current checkpoint

Prompt
Unconditional | target SG

Sample
Qwen autoregressive
temperature: 0.7 → 0.9
→ 1.1 → 1.3

Wyckoff text
Generated candidate

Decode + relax
Parser / PyXtal
eqV2 energy estimate

Build preferences
Stability tiers
Novelty vs. reference set
Target SG match

DPO update
Preferred / rejected text pairs
reference policy: π_{i-1}

Updated policy π_i → next iteration

eqV2 scores relaxed candidates to construct preference pairs; DPO updates the language model from those pairs.

Unconditional and space-group-conditioned pairs are optimized by the same policy.

C Final evaluation

Final policy π^*
Generate samples

Decode structures
Wyckoff text → CIF

eSEN relaxation + evaluation
Stable · Unique · Novel (S.U.N.) | conditional: S.S.U.N.

DFT validation on 1,000-sample subset

Reported outcomes
Higher stability and S.U.N. than the compared baselines
Validation and DFT results are evaluation, not DPO feedback